LEARNING SESSION S9

Interview Questions & Answers

68 questions with model answers: HR, Excel, SQL, statistics, Python, Power BI/Tableau, ML, GenAI, case studies and guesstimates

AnalytixLabs · Advanced Certification in Data Science & AI (IIT), June 2026 batch

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🎯 Interview Questions & Answers

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Learning session · for Data Analyst, Business Analyst, BI Developer and junior Data Scientist roles

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How interviews work



A typical analytics hiring process.

- **Resume**: one page; each bullet = action + tool + measurable result ("Built a Power BI dashboard tracking 12 KPIs, cutting weekly reporting from 6 hours to 30 minutes").
- **Portfolio**: 3 projects on GitHub/Tableau Public/Netlify: one SQL, one Python, one dashboard, each with a business question, approach, and result.
- **Technical round**: live SQL (most common), Excel tasks, Python/pandas, statistics questions.
- **Case round**: business problem → metrics → analysis plan → recommendation.
- **Behavioural**: STAR method (Situation, Task, Action, Result) with numbers.

HR & behavioural

Q Tell me about yourself.

Present → past → future in 60–90 seconds: "I'm a [role] with [x] years in [domain]. Recently I [project with a number]. Before that I [relevant experience]. I'm now moving into data analytics because [reason], and I've built [projects/certification]. This role fits because [company link]."

Q Why data analytics?

Connect a real moment: "In operations I saw decisions made on gut feel; when I built a simple Excel tracker, call volume forecasting improved and we staffed better. I want to do that full-time with stronger tools."

Q Describe a time you found an insight that changed a decision.

STAR: Situation (rising complaints), Task (find the cause), Action (pivot by issue + Pareto showed 3 issues = 70%), Result (process fix reduced complaints 25% in a quarter).

Q How do you handle a stakeholder who disagrees with your numbers?

Walk through definitions and sources together, reconcile differences on a small sample, document the agreed definition, and fix the metric dictionary.

Q How do you prioritise multiple urgent requests?

Clarify business impact and deadlines, agree priority with the manager, communicate timelines, and automate recurring requests.

Q What is your biggest weakness?

A real, improving one: "I used to over-polish reports; now I share a draft early to confirm direction, then refine."

Excel

Q VLOOKUP vs INDEX/MATCH vs XLOOKUP?

VLOOKUP: right-only, column-number fragile. INDEX/MATCH: any direction, robust. XLOOKUP: any direction, built-in not-found and match modes, returns arrays; Excel 365/2021.

Q How do you remove duplicates while keeping the latest record?

Sort by date descending, then Data → Remove Duplicates on the key; or Power Query: sort, then Remove Duplicates (keeps first).

Q Count unique customers who bought in Pune.

`=COUNTA(UNIQUE(FILTER(Cust, City="Pune")))` (365) or a pivot with "Add to Data Model" → Distinct Count.

Q What is a pivot table's "Show Values As"?

Displays values as % of total/row/column, difference from previous, running total, rank, etc., without formulas.

Q Explain absolute vs relative references with an example.

`=B2*$F$1` copied down multiplies each row's B by the fixed rate in F1.

Q How do you handle #N/A from lookups?

Find the cause (trailing spaces, number stored as text, missing key), clean with TRIM/VALUE, and show a friendly value with IFNA or XLOOKUP's if_not_found.

Q What is Power Query and why use it?

Excel's ETL tool: records cleaning steps once and replays them on refresh; combines files and sources.

Q SUMIFS with an OR condition?

`=SUM(SUMIFS(Amt, City, {"Pune","Mumbai"}))`.

Q How do you build a dynamic chart?

Base it on a Table or a FILTER/INDEX range driven by a drop-down; the chart updates automatically.

Q Goal Seek vs Solver?

Goal Seek changes one input to hit one target; Solver optimises an objective with multiple variables and constraints.

SQL

Q Order of execution of a SELECT?

FROM/JOIN → WHERE → GROUP BY → HAVING → SELECT → DISTINCT → ORDER BY → TOP/OFFSET.

Q Find the 3rd highest salary in each department.

```sql
SELECT * FROM (SELECT dept, name, salary, DENSE_RANK() OVER (PARTITION BY dept ORDER BY salary DESC) r FROM Emp) t WHERE r = 3;
```

Q Customers who ordered in January but not February.

```
SELECT customer_id FROM Orders WHERE order_date >= '2026-01-01' AND order_date < '2026-02-01'
EXCEPT
SELECT customer_id FROM Orders WHERE order_date >= '2026-02-01' AND order_date < '2026-03-01';
```

Q Running total of sales by date.

```
SELECT sale_date, amount, SUM(amount) OVER (ORDER BY sale_date ROWS UNBOUNDED PRECEDING) AS running_total FROM Sales;
```

Q Month-over-month growth.

```
WITH m AS (SELECT DATETRUNC(MONTH, order_date) mth, SUM(amount) rev FROM Orders GROUP BY DATETRUNC(MONTH, order_date))
SELECT mth, rev, 100.0*(rev - LAG(rev) OVER (ORDER BY mth)) / NULLIF(LAG(rev) OVER (ORDER BY mth),0) AS mom_pct FROM m;
```

Q Delete duplicates keeping one row.

```
WITH d AS (SELECT *, ROW_NUMBER() OVER (PARTITION BY email ORDER BY id) rn FROM Users)
DELETE FROM d WHERE rn > 1;
```

Q Employees earning more than their manager.

```
SELECT e.name FROM Emp e JOIN Emp m ON e.manager_id = m.emp_id WHERE e.salary > m.salary;
```

Q Percentage of total revenue by category.

```
SELECT category, SUM(amount) rev, 100.0*SUM(amount)/SUM(SUM(amount)) OVER () AS pct FROM Sales GROUP BY category;
```

Q Users active 3 consecutive days.

```
WITH d AS (SELECT DISTINCT user_id, CAST(login_time AS DATE) dt FROM Logins),
g AS (SELECT user_id, dt, DATEADD(DAY, -ROW_NUMBER() OVER (PARTITION BY user_id ORDER BY
dt), dt) grp FROM d)
SELECT DISTINCT user_id FROM g GROUP BY user_id, grp HAVING COUNT(*) >= 3;
```

Q INNER vs LEFT join row counts: when can LEFT return more rows than the left table?

When the right table has multiple matches per key (one-to-many), each left row repeats for every match.

Q CTE vs subquery vs temp table vs view?

CTE: readable, one statement. Subquery: inline. Temp table: stored for the session, indexable. View: saved query object in the database.

Q What is normalisation?

Organising tables so each fact is stored once (1NF, 2NF, 3NF), reducing redundancy and update errors.

Statistics

Q Mean vs median: when to use which?

Median for skewed data or outliers (income, delivery time); mean for symmetric data.

Q What is a p-value?

The probability of results at least this extreme if H0 were true. Below α (0.05) → reject H0. It is not the probability H0 is true.

Q Type I vs Type II error?

Type I: false positive (reject a true H0). Type II: false negative (miss a real effect).

Q Explain a 95% confidence interval.

A range built so that 95% of intervals made the same way would contain the true value.

Q What is the Central Limit Theorem?

Sample means from large enough samples are approximately normal, centred on the population mean with standard error σ/√n.

Q How would you design an A/B test for a new checkout page?

Define the primary metric (conversion) and guardrails (AOV, errors); compute sample size for the minimum detectable effect; randomise users 50/50; run full weeks; test with a two-proportion z-test; check for novelty effects.

Q Correlation vs causation?

Correlation is association; causation needs experiments or causal methods to rule out confounders.

Q Which test compares average spend across 4 regions?

One-way ANOVA (Kruskal–Wallis if data is very skewed/small).

Q What is standard deviation in plain words?

The typical distance of values from the average.

Q Simpson's paradox?

A trend in combined data reverses within subgroups because of a confounding variable (e.g. overall conversion falls while every channel improves, due to a mix shift).

Python

Q loc vs iloc?

loc by labels/boolean masks (end inclusive); iloc by integer positions (end exclusive).

Q apply vs vectorisation?

Vectorised column operations are much faster; use apply only when logic can't be vectorised.

Q How do you merge two DataFrames like VLOOKUP?

`df.merge(lookup, on="key", how="left")`; check row counts before and after.

Q Group and aggregate multiple metrics.

```
df.groupby("city").agg(orders=("id","nunique"), revenue=("amount","sum"))
```

Q Handle outliers?

Detect with IQR or z-score; investigate; cap (clip), transform (log) or remove with justification.

Q What is a lambda?

A small anonymous one-line function, e.g. `lambda x: x * 1.18`.

Q Difference between a list and a NumPy array?

Arrays hold one data type and support fast vectorised maths; lists hold anything and need loops.

Q How do you read only some columns of a big CSV?

`pd.read_csv(f, usecols=[...], dtype={...}, chunksize=100000)`.

Power BI & Tableau

Q Measure vs calculated column (Power BI)?

Columns are stored per row at refresh; measures compute at query time in filter context.

Q What does CALCULATE do?

Evaluates an expression with modified filter context; the core of DAX.

Q Star schema?

Fact table linked to dimension tables with one-to-many relationships; simple, fast, predictable.

Q Row-level security?

Roles with DAX filters (static or USERPRINCIPALNAME-based) assigned to users in the service.

Q Import vs DirectQuery?

Import caches data (fast, needs refresh); DirectQuery queries live (current, slower).

Q Tableau LOD expressions?

FIXED/INCLUDE/EXCLUDE set the aggregation level independent of the view.

Q Tableau context filter?

A filter applied before Top N and FIXED LODs, also improving performance.

Q How do you design a dashboard for executives?

Start from their decisions; 4–6 KPIs with targets and trends at the top; drill-downs below; one message per page; consistent colours; mobile layout.

ML & GenAI

Q Supervised vs unsupervised learning?

Supervised uses labelled outcomes (predict churn); unsupervised finds structure without labels (segments).

Q Bias–variance trade-off?

Simple models underfit (high bias); complex models overfit (high variance); choose complexity by validation error.

Q Precision vs recall?

Precision: of predicted positives, share correct. Recall: of actual positives, share found.

Q How do you handle class imbalance?

Stratified splits, class weights, resampling (SMOTE), threshold tuning, and metrics like recall, F1, PR-AUC.

Q Why scale features?

Distance-based (KNN, k-means) and gradient-based models are sensitive to feature scale.

Q Random forest vs gradient boosting?

RF averages independent trees (robust, less tuning); boosting builds trees sequentially to fix errors (often more accurate, needs tuning).

Q What is cross-validation?

Splitting training data into k folds, training on k−1 and validating on 1, rotating, to estimate performance reliably.

Q What is an LLM hallucination and how do you reduce it?

A confident false output. Reduce with RAG/grounding, asking for sources, lower temperature, and verification.

Q What is RAG?

Retrieve relevant documents by embedding similarity and give them to the LLM to answer from.

Q How have you used GenAI in analytics?

Example: classify 10,000 free-text complaints into themes with an LLM, validate on a 300-row hand-labelled sample (92% agreement), then trend themes in Power BI.

Case & guesstimates

Case framework

1. Clarify the goal and the metric (what does "success" mean?).

2. Break the metric into drivers (revenue = users × conversion × AOV).

3. Hypothesise causes; say which data would confirm each.

4. Analyse by segments: time, channel, region, product, new vs repeat.

5. Recommend, quantify impact, mention risks and next steps.

Q Sales dropped 15% last month. How do you investigate?

Confirm data correctness; check seasonality/YoY; decompose into traffic × conversion × AOV; segment by region, channel, category, new vs returning; check external events (competitor sale, stock-outs, app outage); find the segment explaining most of the drop; recommend actions.

Q Which metrics would you track for a food-delivery app?

North star: orders delivered per week. Drivers: active users, order frequency, conversion, AOV, delivery time, cancellation rate, restaurant availability, rating; guardrails: refunds, delivery partner earnings.

Q Guesstimate: daily UPI transactions in a city of 1 crore people.

Adults with smartphones ~60% → 60 lakh; UPI users ~70% → 42 lakh; average 1.5 transactions/day → ~63 lakh transactions/day. State assumptions clearly; the structure matters more than the number.

Q How would you measure whether a new feature worked?

A/B test with a pre-defined primary metric and guardrails; if not possible, difference-in-differences against a comparable group.

Resources

- ▶ YouTube: Simplilearn data analyst interview questions and answers · ▶ YouTube: Simplilearn SQL interview questions · ▶ YouTube: Simplilearn Power BI interview questions
- ▶ YouTube: techTFQ SQL interview questions · ▶ YouTube: Alex The Analyst data analyst interview
- 📘 DataLemur (SQL and statistics interview practice) · 📘 StrataScratch · 📘 LeetCode SQL 50
- Course-specific Q&A also sits in the Interview Q&A tab of every module and session.